

Big data and analytics for crop yield forecasting: empirical research and development prospects

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Abstract. The study provides a comprehensive analysis of the role and prospects of applying big data technologies in agriculture. It covers a wide range of issues related to the implementation and use of Big Data in the agro-industrial sector, exploring both the theoretical foundations and practical aspects of their application. Special attention is given to the examination of current trends, identification of key challenges, and opportunities associated with the use of these technologies in agriculture. The authors investigate how big data technologies are transforming approaches to managing agrarian processes, improving crop yields, and optimizing resources. Various aspects are analyzed, including the development of data processing technologies, their application for analysis and forecasting in agriculture, and discussions on issues related to the adoption and dissemination of these technologies in the Russian context. Specific examples of successful projects and initiatives demonstrating the potential of Big Data in agribusiness are presented.

1 Introduction

Modern business is in the era of digital transformation, where the generation and analysis of vast amounts of data have become crucial for making effective managerial decisions. The increasing unstructured nature of data demands faster processing speeds, surpassing the capabilities of traditional business analytics systems. In this context, Big Data technologies offer a solution for working with both structured and semi-structured, as well as unstructured data. The key characteristics of Big Data include volume (large data sets), velocity (high speed of data generation and processing), and variety (data diversity).

The rapid growth in the popularity and effectiveness of Big Data technologies is driven by several IT factors, including advances in computational hardware performance, an increase in the number of processor cores, a reduction in the cost of storage resources, and

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the widespread availability of cloud services. A significant contribution to the development of this field has been made by the Hadoop platform with its distributed computing MapReduce and distributed file system HDFS, enabling efficient storage and processing of large volumes of diverse data.

The specificity of agriculture, as one of the key sectors of the economy, requires a special approach in the application of big data technologies. The industry has a need for optimizing production processes, improving crop yield, and resource management, where the use of Big Data opens up new horizons. Examples of successful projects, including Sber's Agro AI, an Australian project for vineyards, the British Intelligent Fruit Vision system, and many others, confirm the significant potential of utilizing data technologies to enhance the efficiency and sustainable development of the agricultural sector.

2 Materials and methods

This article utilized materials sourced from various outlets, including academic publications, reports from research institutions, and data provided by industry experts and companies operating in the fields of agrotechnology and big data. To analyze the current state and future prospects of implementing big data technologies in agriculture, a comprehensive approach was employed, encompassing both qualitative and quantitative data analysis.

Qualitative analysis encompassed the examination and systematization of literature on the subject of big data, the analysis of reports and studies related to the implementation and utilization of these technologies in the agricultural sector. Special attention was given to studying practical cases and projects implemented in various countries, allowing for the assessment of real-world examples of Big Data technology application in agriculture and identifying key factors contributing to their effectiveness.

Quantitative analysis was conducted based on statistical data pertaining to the volume and structure of the big data market, as well as the level of their integration into various spheres of the agro-industrial complex. Data provided by government institutions, research institutes, and industry associations were utilized.

The study adhered to the principles of objectivity and a comprehensive approach, ensuring a deep understanding of the subject matter and its practical applicability in the real conditions of agricultural production.

3 Results

In recent years, the application of big data technologies has actively penetrated the field of agriculture, opening new possibilities for improving efficiency and crop yield. This process traces its roots back to the first half of the 20th century when a systematic approach to forecasting crop yield was developed, based on the advancement of agrometeorology and knowledge of plant physiology.

The mid-20th century marked the emergence of crop yield forecasting methods based on agrometeorological observations and temperature characteristics during the period of maximum plant biomass growth. These methods were used to calibrate mathematical models, varying calculation methods and considering various factors influencing crop yield.

The main challenge in crop yield forecasting is the multifactorial and heterogeneous nature of the system: the growth and development of plants are influenced by both biotic and abiotic factors, each exerting its unique impact. Plants respond to numerous conditions, including the physical and chemical properties of the soil, temperature indicators, light

levels, precipitation intensity and quantity, as well as fluctuations in soil moisture reserves and the influence of microorganisms.

Accurate quantitative consideration of each of these factors requires regular direct or indirect measurements, which poses a labor-intensive task in the conditions of industrial crop cultivation. Despite the evolution of means and methods of measurement, direct investigation of all factors still does not provide the possibility of a comprehensive analysis for each individual enterprise and field [1].

As a result, mathematical models based on direct measurements of a wide range of indicators have not gained widespread adoption in real production conditions. However, these methods remain relevant in experimental science and may find broad application in the future when simpler and more efficient ways of accurately measuring influencing factors on individual fields are discovered.

During the implementation of crop yield forecasting technologies in agriculture, many specialists encounter questions about the reasons for discrepancies between forecasted data and actual crop yields. To understand these differences, it is important to consider typical situations that impact the accuracy of forecasts [2]:

- **Crop Contamination:** Vegetation indices that monitor the development of green biomass do not differentiate between crops and weeds. This means that in highly contaminated fields, forecasted values may significantly deviate from actual ones. Despite the existence of technologies to distinguish weeds from crops in satellite images, they do not always provide absolute accuracy, especially when weeds and crops belong to the same botanical type.
- **Field Heterogeneity:** Cases often arise where only a portion of a field is sown due to various factors such as excessive moisture or erosion development. This can lead to inaccuracies in forecasts since forecasting programs may assume that the entire field is sown with the same crop. Such situations require adjustments to digital field contours for accurate forecasting.
- **Morphology of Varieties and Hybrids:** The diversity of varieties and hybrids within a single crop species can result in significant morphological differences affecting forecast accuracy. New varieties or hybrids may temporarily increase forecasting errors until sufficient information is obtained for proper algorithm calibration.
- **Integration of Forecasting into Enterprise Management:** Crop yield forecasting functions become an essential part of digital farm management systems. They enable the optimization of harvest plans, assessment of storage resources, and the making of effective managerial decisions based on predicted yield data.

It is important to note that remote sensing-based forecasting methods are most effective for grain crops. However, when predicting the yield of crops with underground biomass, such as sugar beets, additional challenges arise due to the discrepancy between above-ground and underground plant parts, leading to increased forecast inaccuracies. In such cases, further refinement of forecasting methods and adaptation to the specific characteristics of individual crops are required.

Yield forecasting methods based on spectral indices typically exhibit a deviation from actual yields of no more than 10%. The process of local algorithm calibration plays a crucial role in achieving such accuracy.

It involves adapting the calculation mechanism based on historical data on crop yield in a specific locality. Considering that most modern digital farm management systems are equipped with crop yield forecasting functions, accumulated data on crops and their yields become a valuable resource. The accuracy of forecasts increases with the "digitalization tenure" of the farm, as well as with a well-documented history of yield for each field.

There are two main methods of crop yield forecasting using vegetation indices: the biomass peak method and the analog year method. The biomass peak method is based on

analyzing the maximum values of the vegetation index at a specific stage of plant development, typically providing maximum forecast accuracy 1-2 months before harvest. The analog year method, applied in the early stages of vegetation, often has lower accuracy with larger time intervals until harvest.

In the context of collecting and processing big data, various software and technological complexes are utilized, combining software, hardware, and services. In the realm of software, key approaches include Database Management Systems (DBMS), NoSQL, NewSQL, Hadoop, In-memory Database, and SAP HANA. Big data manufacturers, such as IBM, SAP, Microsoft, Oracle, offer a wide range of solutions, ranging from data storage and preprocessing to the development of sophisticated analytics algorithms [3].

In agriculture, the use of big data opens new perspectives for increasing crop yield. This is achieved through the analysis of numerous factors influencing yield, such as meteorological conditions, cultivation methods, and irrigation parameters. It is believed that the application of analytical services based on big data can significantly enhance the yield of grain crops and improve milk production, provided correct managerial decisions are made based on data analysis.

In the field of crop cultivation, the use of big data occurs through two main directions: services offered by major companies and software for independent use by clients. Companies provide comprehensive services, including analytical support for the crop cultivation process. They utilize their own data (regional weather conditions, seed germination, and yield) and customer data (field location, historical yield, soil type), as well as information from field devices (crop distribution, planting density). Based on this analysis, recommendations are formulated for agricultural producers. The entire crop cultivation process is automated, and company specialists provide technical support throughout the subscription period.

An alternative is independent use of big data software. Farmers input information about harvest collection, sensor data, and laboratory data into the system, integrating it with the state information system. This allows the program to analyze information and propose optimal farm management strategies.

Various solutions for data collection, processing, and analysis are present in the Big Data market. These developments encompass specialized equipment (sensors for soil moisture determination, predictive weather stations, devices for monitoring agricultural machinery) and software that processes and analyzes the collected data, providing information in a user-friendly format. Programs can collect data directly for the client or onto cloud servers, where information is processed and presented to the client in the form of ready-made analytics.

Big data analytics in agriculture contributes to the timely identification of field issues, reducing the need for manual labor. Specialized programs use data on weather conditions, soil, and plant status to assist farmers in making decisions on optimal planting and harvesting times, irrigation methods, and fertilizer dosages. This increases the efficiency of agricultural production, reducing costs and optimizing management processes [4].

Among the big data analysis methods in demand in agriculture are intelligent data analysis, machine learning, pattern recognition, predictive analytics, spatial analysis, and time series analysis. The implementation of these methodologies has become possible thanks to modern analytical tools. According to a study by ISSEZ NRU HSE, based on a survey of over 1,700 organizations in Russia, 45% of IT companies are actively involved in developing solutions based on big data technologies, with an additional 7% planning to start in the next five years. International products and solutions based on open-source technologies dominate the Russian market, with active participation from companies such as CleverDATA, "Fors - Development Center," AT Consulting, and others.

The Russian government sector also makes a significant contribution to the development and application of Big Data technologies. Since November 2021, the first national standard in the field of big data has been in effect - GOST "Information Technologies. Big Data. Overview and Vocabulary," developed by the National Center for Digital Economy at Lomonosov Moscow State University and the Institute for the Development of the Information Society. This document creates a terminological basis for standards related to big data and anticipates the adoption of additional standards aimed at improving the application of technologies in Russia [5].

Existing educational projects, such as the "Yandex School of Data Analysis," courses from Netology, and Coursera, also contribute to the dissemination of knowledge about working with big data. These courses educate professionals on the use of analytical tools and methodologies, which is crucial for the development of the digital economy in Russia.

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In 2020, according to statistics, 22.4% of organizations in Russia utilized big data collection, processing, and analysis technologies, with this figure standing at 17.2% in the agricultural sector. However, the adoption of Big Data analytical tools in Russia remains lower than in developed countries, reflecting the need for further development and adaptation of these technologies. The effective use of big data technologies requires programs that are not only user-friendly but also ensure data protection and a return on investment.

Therefore, it can be noted that Big Data in agriculture represents a promising direction that combines innovation, government support, and active participation from the private sector, contributing to the increased efficiency and productivity of the industry.

In agriculture, numerous innovative projects based on big data are being implemented, facilitating the optimization of production processes and resource management. One such example is the Damate Group, actively utilizing Big Data and predictive analytics. The company has implemented online monitoring in livestock processes and developed the MathLab laboratory for big data analysis and machine learning, optimizing the production and sales processes.

The company City Farmer is also implementing digital technologies by creating remotely controlled farms based on artificial intelligence. These innovations are applied in various fields, from growing greens and mushrooms to egg and fish production.

The agro-holding AFG National has implemented digital technologies, including field monitoring using drones, installation of field battery-powered weather stations, and the development of digital work and expenditure accounting systems. The Agricultural Company Rus applies analytics and forecasting services based on AI for dairy production.

The company Step has developed its own software for predicting apple yields using neural networks and introduced the cloud service "Field History" for data collection and analysis, optimizing crop rotation and assessing market demand.

The EcoNiva Group has introduced the EkoCrop application for crop farming, providing data collection and analysis for field operations. Additionally, the company utilizes RFID animal identification and proprietary programs for optimizing the feeding process.

The Agro AI project by Sber is an example of advanced application of artificial intelligence in agriculture. This unique system can assess fields, forecast yields, and conduct in-depth analysis of crops using a comprehensive approach with the use of drones, robots, and AI. The development of this project, carried out entirely by Sber and SberAnalytics specialists, underscores the significant potential of IT technologies in agriculture.

An Australian project for vineyards also demonstrates innovative technology use in the agricultural sector. Queensland University experts proposed a system using UAVs for monitoring vineyards. Tested in 2016, this system proved highly effective in yield assessment, which is crucial for Australia as a major grape producer.

The British company Intelligent Fruit Vision developed the Fruitvision system for monitoring fruit and vegetable crops, providing accurate yield data. This comprehensive solution includes automated systems, robots, and UAVs, allowing for more efficient management of agricultural production.

These examples demonstrate how the integration of big data and digital technologies into agriculture contributes to improving production process management, increasing efficiency, and automating key operations [6].

Specialists from Rosselkhozbank highlight several key trends that will shape the development of the Russian agro-industrial complex (APK) until 2025. Among them are the development of proprietary seeds, the creation of organic fertilizers and plant protection agents, the advancement of digital services based on precision farming, and the transition of agriculture to online trading formats. Remarkably, crop forecasting through satellite imagery is one of the boldest and most promising directions. The development of the space era and Earth remote sensing systems opens new horizons for precise monitoring of agricultural lands, increasing the efficiency of forecasting and management in agriculture [7].

These examples illustrate how innovations in Big Data, AI, and space technologies contribute to a revolutionary transformation in the agro-industrial complex, unlocking new possibilities for increased productivity and management efficiency.

In terms of more ambitious development plans, the forecasting of crop yields using satellites deserves special mention. The accessibility of satellite imagery of agricultural lands obtained in various spectral ranges has revealed extensive opportunities for agriculture. These data not only allow for the differentiation of vegetation cover based on biomass levels, using indicators such as NDVI and other difference indices, but also serve as the foundation for creating innovative crop yield forecasting models. Research initiated in the 1980s revealed that changes in spectral indices during the vegetation process of agricultural crops correlate with yield levels.

The correlation relationship between vegetation indices and crop yield volume varies depending on the biological characteristics of different crops. Overall, a direct correlation has been observed between vegetation index values and final production, whether it be grain or green biomass.

Vegetation indices, thus, serve as an integral metric in a multifactorial system, measuring biomass as a function of multiple influencing variables. This discovery significantly enhances the potential of precision agriculture, providing the opportunity for more accurate crop yield forecasting and efficient resource management based on remote sensing data.

The use of satellite monitoring data in agriculture is, therefore, one of the key directions in the application of big data technologies. This enables farmers not only to accurately assess the condition of crops and make timely decisions regarding crop care but also to effectively forecast potential harvests, improving planning and resource management. Such an approach contributes to increased crop yields, reduced costs, and the optimization of agricultural production, which is a crucial factor in modern agribusiness.

4 Discussion

The application of big data technologies in Russian agriculture, despite its significant potential, faces several restraining factors. One key factor is the shortage of qualified professionals capable of implementing Big Data projects. This issue is particularly relevant when educational programs have not fully adapted to the rapidly changing requirements of the digital economy.

The high cost of commercial big data processing tools is also a significant barrier. Most solutions available in the market are provided by international companies, increasing the financial burden on domestic enterprises. Additionally, the complexity of integrating new data processing systems with existing software at enterprises requires additional time and financial investments.

The lack of practical experience among most integrators and manufacturers, coupled with customers' concerns about data confidentiality, reinforces consumer mistrust in Big Data technologies. These factors hinder the broader application of these technologies in Russian agriculture.

Nevertheless, rational and efficient use of big data is one of the most powerful tools for attracting investments and ensuring economic growth. Considering the growing need to optimize agro-industrial production and resource management, Big Data technologies can play a key role in increasing crop yields and the efficiency of the agricultural sector. This is supported by a series of successful projects implemented both in Russia and abroad, demonstrating the significant potential of using these technologies in the agricultural sector.

Taking these factors into account, it is important to focus on the development of educational programs related to big data processing, strengthening internal competencies, and contributing to the development of domestic tools and solutions in this field. This will help reduce barriers to the implementation of big data technologies and accelerate the process of digitizing Russian agriculture.

5 Conclusion

The conducted research leads to the conclusion that big data technologies play a critically important role in modern agriculture, providing powerful tools for improving resource management, increasing crop yields, and optimizing production processes. Through the

analysis of large volumes of diverse data, agricultural enterprises can make more informed decisions based on accurate and up-to-date information.

The main obstacles to broader adoption of big data technologies in Russia include the lack of qualified professionals, high costs of commercial tools, integration complexities with existing systems, and concerns about data confidentiality. However, with the right approach and overcoming these barriers, the potential of big data technologies can be fully realized, leading to significant economic growth and innovative development of the agro-industrial complex.

Successful examples of big data application in agriculture, both in Russia and other countries, underscore the significance of these technologies for the future of the agricultural sector. The implementation of innovative solutions based on big data analysis opens new opportunities for increasing crop yields, efficient land resource management, and optimizing production processes, ultimately contributing to the sustainable development of agriculture.

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